

Optical Coherence Tomography (OCT) & Machine Learning — Technical Overview

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OCT is a non-invasive, micron-resolution imaging method that reconstructs cross-sectional and volumetric pictures of tissue from back-scattered near-infrared light. In ophthalmology it is the standard of care for the retina, and it has become one of the most active testbeds for deep learning in all of medical imaging — with strong headline accuracy on curated benchmarks, but persistent gaps in cross-device generalization, robustness, and trustworthy deployment.

What OCT is: the underlying physics

OCT is the optical analogue of ultrasound: instead of timing sound echoes, it measures the echo delay and intensity of near-infrared light back-scattered from tissue. Because light travels far too fast to time directly, OCT recovers depth through **low-coherence interferometry** — a Michelson interferometer splits broadband light into a sample arm (into the eye/tissue) and a reference arm (a mirror), and interference between the two beams only occurs when their path lengths match to within the light's short coherence length. Sweeping or resolving that match reconstructs a depth-resolved reflectivity profile.

The hierarchy of what OCT produces:

- **A-scan** — a single depth (axial) reflectivity profile at one lateral point, analogous to a core sample.
- **B-scan** — a 2D cross-section built from many adjacent A-scans; this is the familiar retinal “slice” that most classifiers consume.
- **C-scan / en-face** — a coronal (frontal) plane extracted at one depth.
- **Volume (cube)** — a stack of B-scans giving full 3D structure; the clinically richest but heaviest data object.

Key physical properties that shape the images and the ML problem:

- **Axial resolution** ($\sim 1\text{--}15\ \mu\text{m}$, typically $3\text{--}7\ \mu\text{m}$ in retina) is set by the source bandwidth, not the focusing optics, and is decoupled from lateral resolution ($\sim 15\text{--}20\ \mu\text{m}$). This is why OCT resolves individual retinal layers that fundus photography cannot.

- **Penetration** is shallow (~1–2 mm in tissue) because light scatters strongly — fine for the transparent eye, limiting elsewhere.
- **Speckle noise** is intrinsic (coherent interference of sub-resolution scatterers), giving a grainy texture that is both signal and nuisance; combined with low signal-to-noise ratio and eye-motion artifacts, it is a first-order concern for any downstream model.

OCT was introduced in 1991 (Huang, Fujimoto et al., *Science*) and matured clinically once Fourier-domain detection made it fast enough for routine volumetric scanning.

Modalities, hardware, and why device matters

The generation of detection technology determines speed, penetration, and — critically for ML — the noise and scaling characteristics that cause domain shift.

- **Time-domain OCT (TD-OCT)** — original design; a physically moving reference mirror scans depth. Slow (hundreds of A-scans/s), now largely historical (e.g. legacy Stratus).
- **Spectral-domain OCT (SD-OCT)** — Fourier-domain; a fixed reference and a spectrometer with a line camera capture the whole depth profile at once. ~800–870 nm, tens of thousands of A-scans/s. The workhorse of retinal imaging and of nearly all public datasets.
- **Swept-source OCT (SS-OCT)** — Fourier-domain with a rapidly tunable laser (~1050 nm) and point detector. Longer wavelength penetrates deeper (choroid, dense cataract), faster still (100k–400k+ A-scans/s), less roll-off. Increasingly standard in newer devices.
- **OCT Angiography (OCT-A)** — not a new hardware class but a processing mode: repeated B-scans at the same location, with motion contrast (decorrelation between frames) revealing blood flow. Produces depth-resolved vascular maps **without dye**, unlike fluorescein angiography.

Scan geometry varies by protocol — single foveal line scans, dense macular cubes, radial/star patterns, RNFL circle scans around the optic disc — and this heterogeneity (slice count, resolution, aspect ratio) is a real obstacle for models that expect fixed-size input.

Vendors and their fingerprints. The main clinical devices are Zeiss (Cirrus), Heidelberg (Spectralis), Topcon (Triton, Maestro), Optovue/Visionix (Avanti, Solix), Nidek, and Canon. Each has distinct axial/lateral resolution, noise profile, image scaling, and proprietary post-processing. A model trained on one vendor routinely degrades on another — the single biggest barrier to clinical adoption (covered under Challenges).

Clinical usefulness

OCT's value is that it is fast, non-contact, and resolves microstructure that no other in-vivo modality shows, enabling early detection, quantitative monitoring, and image-guided treatment.

Retina (the dominant use). OCT is the standard of care for macular disease. It visualizes and quantifies:

- **Age-related macular degeneration (AMD)** — drusen (dry), choroidal neovascularization/CNV and geographic atrophy (late); OCT drives anti-VEGF treatment decisions.
- **Diabetic macular edema (DME) and diabetic retinopathy** — retinal thickening and fluid, a leading cause of working-age blindness.
- **Retinal fluid** — intraretinal fluid (IRF), subretinal fluid (SRF), and pigment epithelial detachment (PED); their presence and volume guide re-treatment intervals.
- Retinal vein/artery occlusion (RVO/RAO), epiretinal membrane (ERM), macular hole, vitreomacular traction, and central serous retinopathy (CSR).

Glaucoma. Quantitative thinning of the retinal nerve fibre layer (RNFL) and ganglion cell complex (GCC) around the optic disc supports diagnosis and progression tracking, often before visual-field loss.

Anterior segment (AS-OCT). Cornea (pachymetry, keratoconus, refractive-surgery planning), anterior chamber angle assessment for angle-closure glaucoma, and intraocular lens sizing.

Beyond the eye. Intravascular OCT images coronary artery walls and stent apposition in cardiology; OCT is also used in dermatology, dentistry, and endoscopic/gastrointestinal imaging — though ophthalmology is where the data volume, benchmarks, and AI activity are concentrated.

Because OCT produces standardized, quantitative, high-throughput images with an unambiguous “ground truth” anatomy, it is unusually well suited to automation — which is why it became a focal point for medical-imaging ML.

The ML / DL / AI task landscape on OCT

Deep learning has been applied across the full OCT pipeline. The main task families:

- **Disease classification / triage** — B-scan or volume-level labelling (e.g. CNV / DME / drusen / normal), and “urgent-referral” triage. The most-published task; where the headline “matches ophthalmologists” claims come from.
- **Layer segmentation** — delineating retinal layer boundaries (NFL, GCL+IPL, RPE, ellipsoid zone, etc.) to derive thickness maps. Dominated by U-Net variants; recent work targets a *universal* segmenter across devices and diseases (e.g. UniOCTSeg).
- **Fluid / lesion segmentation** — pixel-level IRF/SRF/PED, drusen, and geographic-atrophy segmentation for quantitative biomarkers. The RETOUCH challenge is the canonical benchmark; nnU-Net and attention-U-Net variants lead.
- **Denoising and super-resolution** — speckle suppression and resolution enhancement, increasingly with diffusion models and self-supervised methods; important because noise both hides pathology and destabilizes downstream models.
- **Biomarker quantification** — automated fluid volume, retinal/RNFL thickness, drusen load, GA area — the outputs clinicians actually act on.
- **Prognosis / treatment-response prediction** — forecasting AMD progression, or predicting anti-VEGF response and re-treatment need from baseline scans. High clinical value, still emerging.
- **Cross-modal synthesis** — generating OCT-like information from cheaper fundus photos, or fundus↔OCT translation (e.g. the APTOS-2024 fundus→OCT challenge), to extend access where OCT hardware is scarce.
- **Anomaly / out-of-distribution detection & uncertainty** — flagging rare or unseen pathology and abstaining when the model shouldn't be trusted. The least mature but arguably most safety-critical family, and the crux of moving from accuracy to reliability.

A structural theme runs through all of these: most published systems operate on individual 2D B-scans, whereas clinicians read the full 3D volume — so a growing frontier is moving models from slices to volumes.

Public datasets and benchmarks

Progress is anchored on a handful of public sets. Note the split between classification sets (B-scan labels) and segmentation challenges (pixel masks), and the recurring problem that many classification sets are small, single-vendor, or leak between train and test.

Dataset	Task	Classes / labels	Scale	Notes
Kermany / OCT2017	Classification	CNV, DME, drusen, normal	~84k B-scans, ~5.3k patients	The default benchmark; near-saturated (~97–99%). Known train/test data-leakage issues inflate reported accuracy (esp. v2).
OCT-C8 / OCT-8C	Classification	8 classes (adds AMD, CSR, MH, DR)	~24k images	Harder multi-class successor to Kermany.
OCTDL	Classification	7 (AMD, DME, ERM, RAO, RVO, VID, normal)	~2k+ labelled images	Newer open set (Optovue), more disease diversity.
OCTID	Classification	AMD, CSR, DR, MH, normal	~500 images	Small; often combined with others.
RETOUCH	Fluid segmentation	IRF, SRF, PED	70 volumes, 3 vendors (Cirrus, Spectralis, Triton)	The canonical fluid benchmark (MICCAI 2017); multi-vendor by design.
Duke SD-OCT	Classification + layer seg	DME/AMD; layer boundaries	Srinivasan (45 vol.), Farsiu/Chiu sets	Long-standing layer-segmentation references.
OLIVES	Multimodal	Clinical labels + biomarkers + fundus	~78k OCT + labels	Pairs OCT with clinical/biomarker data for richer tasks.
UMN	Fluid	Exudative AMD	~600 B-scans	Single-disease fluid set.

Dataset	Task	Classes / labels	Scale	Notes
Glaucoma sets (GAMMA, GOALS, etc.)	Classification / seg	Glaucoma, optic-disc/cup	Varies	Fundus+OCT and OCT-specific optic-nerve tasks.

Recent challenges keep raising the bar: RETOUCH (fluid), APTOS-2024 (fundus→OCT synthesis), and the MARIO challenge (2024–2025, AMD progression / lesion evolution) push toward volumes, cross-modality, and longitudinal tasks rather than single-slice classification. A 2025 multimodal OCT foundation-model **benchmark** (Morano et al., *npj Digital Medicine*) is an attempt to standardize comprehensive evaluation across tasks.

State of the art: from CNNs to foundation models

The field has moved through three broad eras.

- 1. Transfer-learned CNNs (2017–~2021).** The Kermany et al. 2018 paper (InceptionV3, transfer learning) set the template: ImageNet-pretrained backbones (VGG16, ResNet50, Inception, EfficientNet, DenseNet) fine-tuned on OCT B-scans, reaching expert-comparable accuracy on curated sets. For segmentation, **U-Net** and then **nnU-Net** (self-configuring) became — and remain — the default; RETOUCH-era winners were almost all U-Net-based.
- 2. Vision Transformers (~2021–2023).** ViT and hybrid CNN-transformer models were shown to match or beat CNNs on OCT classification and to offer different robustness/attention properties, though they are data-hungry and don't automatically generalize better out-of-distribution.
- 3. Foundation models & self-supervised pretraining (2023–now)** — the current frontier:
 - **RETFound** (Zhou et al., *Nature* 2023) — a masked-autoencoder ViT self-supervised on ~1.6M retinal images (OCT + fundus). The reference retinal foundation model: fine-tunes to many downstream tasks with strong label efficiency and generalization, and is now the standard baseline others benchmark against.
 - **3D / volume-native models** — e.g. **OCTCube-M** (3D multimodal, cross-device/cross-cohort validation) and V-JEPA-style volume pretraining, which shift the paradigm from B-scan slices to full volumes and report significant AUROC gains for AMD and glaucoma over image-based models.

- **Multimodal & vision-language models** — EyeFound, VisionFM, RetiZero, FLAIR and MM-Retinal integrate OCT with fundus, text/report supervision, and clinical data; a 2025 multimodal OCT foundation model + benchmark (Morano et al.) targets comprehensive analysis in one model.
- **Robustness-oriented architectures** — work like FlexiVarViT processes B-scans at native resolution and variable slice counts to preserve information and improve external-set robustness, reflecting a shift in emphasis from peak accuracy to generalization.

The throughline: pretraining scale and self-supervision now matter more than the specific backbone, and the research center of gravity has moved from “which CNN classifies Kermany best” to “which pretrained representation transfers and holds up on unseen devices and populations.”

Representative benchmark numbers

Headline metrics are strong but must be read with two caveats: (1) high accuracy on **internal** curated splits routinely drops on **external** data, and (2) some classic numbers are inflated by train/test leakage. Figures below are representative, not a leaderboard.

Task	Dataset	Model	Metric	Result
4-class classification	Kermany/OCT2017 (internal)	InceptionV3 (Kermany 2018)	Accuracy / AUC	~96.6% / 0.999
4-class classification	Kermany	ResNet50 / VGG16	Accuracy	~0.998 (near-saturated)
7-class classification	OCTDL	VGG16 / ResNet50	Accuracy	~0.86 / 0.85
5-class classification	OCTID	VGG16	Accuracy	~0.93
Cross-dataset (train→unseen)	Kermany + OCTDL	ResNet50	Accuracy	~0.83 (drop vs internal)
Fluid segmentation	RETOUCH	3D encoder-decoder	Dice (Cirrus)	~0.73

Task	Dataset	Model	Metric	Result
Fluid segmentation	RETOUCH	Attention U-Net variant	F1 (IRF / SRF / PED)	0.82 / 0.93 / 0.94
AMD & glaucoma detection	5 OCT sets	V-JEPA (volume) vs best image model	mean AUROC	0.94 vs 0.90
Referral triage	Kermany-style	Ensemble ResNet50	Sensitivity / specificity	~0.96 / ~0.99

How to read this. On clean single-vendor B-scan classification, the problem is close to solved (>97%). The real, unsolved gap is the ~10–15 point degradation when models meet a new device, protocol, or population — which is exactly where headline numbers stop predicting clinical behaviour. Segmentation Dice for the hardest fluid type (IRF) still trails, and even RETOUCH found that *fusing* methods beat any single one, i.e. individual segmentation is not yet at inter-grader reliability.

Open challenges (research and industry)

The gap between benchmark accuracy and dependable clinical deployment is where the real problems live.

- **Cross-vendor / cross-protocol domain shift.** The single biggest barrier to adoption. Devices differ in resolution, noise, scaling and proprietary processing, so a model trained on one vendor degrades on another; real clinics run multiple vendors and upgrade over time. Mitigations — domain adaptation, adversarial/CycleGAN style transfer, noise-transfer augmentation, native-resolution processing — help but don't close the gap.
- **Benchmark leakage and over-optimistic metrics.** Widely used sets (notably Kermany v2) have train/test overlap at the patient level, inflating reported accuracy. Patient-level splitting and external validation are still inconsistently applied, so published numbers often overstate real performance.

- **Robustness, uncertainty, and knowing when not to trust the model.** Systems are typically confident even when wrong, and rare high-stakes pathologies are exactly the tail they were undertrained on. Out-of-distribution detection, calibrated uncertainty, and selective prediction (abstaining/deferring to a clinician) are immature but arguably the most safety-critical direction — accuracy without reliable abstention is not clinically safe.
- **Data scarcity, annotation cost, and imbalance.** Pixel-level fluid/layer masks require scarce expert time; rare diseases are underrepresented; privacy limits data sharing. This drives the pull toward self-supervised and foundation-model pretraining.
- **2D slices vs 3D volumes.** Most models read single B-scans while clinicians read the cube; volume-native models are heavier and less mature but capture context a slice can't.
- **Interpretability and clinician trust.** Saliency/attention maps are often unstable and not anatomically faithful; structure-aware, anatomy-aligned explanation is an active area. Without trustworthy explanations, uptake stalls.
- **Regulatory, reimbursement, and prospective evidence.** Autonomous AI exists (the first FDA-cleared autonomous diagnostic, LumineticsCore/IDx-DR, 2018; also EyeArt, AEYE, Retina-AI Galaxy — mostly fundus-based DR screening), and CPT code 92229 (2021) enabled US reimbursement. But most OCT models never reach prospective, real-world validation; internal testing alone is insufficient for clearance.
- **Fairness and equity.** Vendor and demographic bias can disadvantage patients served by older or less common devices; representative external validation is needed to avoid encoding inequities.
- **Human-AI workflow integration.** Evidence increasingly favours collaborative (assistive) workflows over fully autonomous quantification, especially for atypical or low-quality scans — the model as a second reader, not a replacement.

Sources

Key references behind the SOTA, benchmark, and challenge sections (as of 22 Sep 2026):

- [RETFound — A foundation model for generalizable disease detection from retinal images \(Zhou et al., *Nature* 2023\)](#)
- [Multimodal foundation model and benchmark for comprehensive retinal OCT analysis \(Morano et al., *npj Digital Medicine* 2025\)](#)
- [Shifting retinal foundation models from slices to volumes / V-JEPA \(*npj Digital Medicine* 2026\)](#)

- [OCTCube-M — 3D multimodal OCT foundation model](#)
- [FlexiVarViT — robust multi-disease OCT classifier \(*Scientific Reports* 2025\)](#)
- [RETOUCH — Retinal OCT Fluid Detection & Segmentation Benchmark \(IEEE TMI\)](#)
- [OCTDL — OCT dataset for image-based deep learning \(*Scientific Data* 2024\)](#)
- [Inflation of test accuracy due to data leakage in OCT classification \(arXiv\)](#)
- [Domain adaptation for multivendor OCT dependence in deep learning \(Zenodo, 2026\)](#)
- [Deep learning for the MARIO Challenge — AMD progression \(arXiv 2025\)](#)
- [APTOS-2024 — synthetic 3D OCT from fundus \(arXiv 2025\)](#)
- [Artificial Intelligence in Ophthalmology \(EyeWiki, 2026\)](#)
- [AI for DR screening: where are we in 2025? \(Retina Specialist\)](#)
- [Integrating human expertise with AI for OCT fluid quantification — systematic review \(*Cureus* 2025\)](#)

Historical: Huang et al., *Science* 1991 (introduction of OCT); Kermany et al., *Cell* 2018 (OCT2017 benchmark).